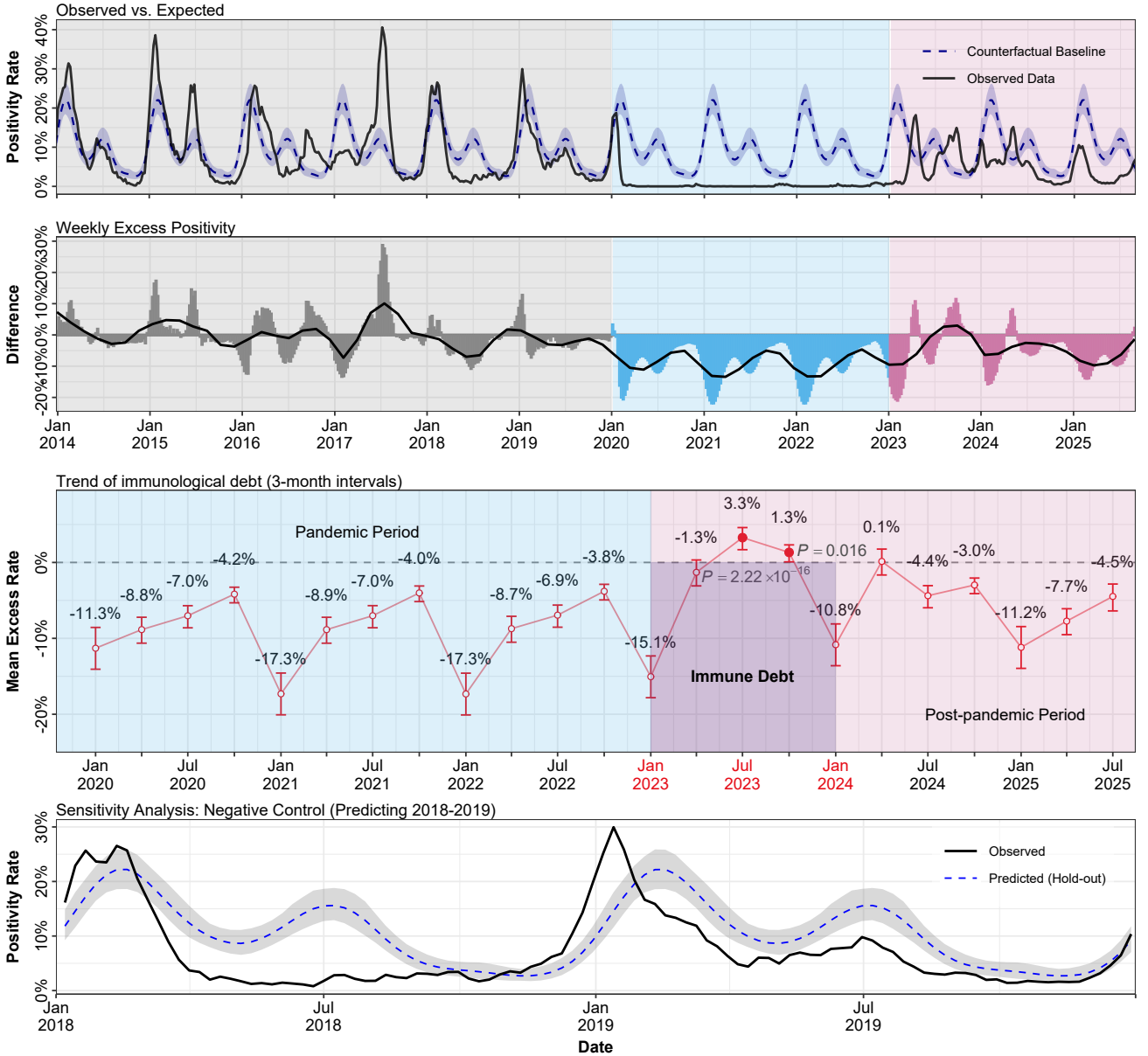

Observed vs. Expected
Positivity Rate
0%
10%
20%
30%
40%
Counterfactual Baseline
Observed Data
Weekly Excess Positivity
Difference
-20%
-10%
0%
10%
20%
30%
Jan 2014
Jan 2015
Jan 2016
Jan 2017
Jan 2018
Jan 2019
Jan 2020
Jan 2021
Jan 2022
Jan 2023
Jan 2024
Jan 2025
Trend of immunological debt (3-month intervals)
Mean Excess Rate
Pandemic Period
Immune Debt
Post-pandemic Period
-11.3%
-8.8%
-7.0%
-4.2%
-17.3%
-8.9%
-7.0%
-4.0%
-17.3%
-8.7%
-6.9%
-3.8%
-15.1%
-1.3%
3.3%
1.3%
P = 0.016
P = 2.22 ×10−16
-10.8%
0.1%
-4.4%
-3.0%
-11.2%
-7.7%
-4.5%
Jan 2020
Jul 2020
Jan 2021
Jul 2021
Jan 2022
Jul 2022
Jan 2023
Jul 2023
Jan 2024
Jul 2024
Jan 2025
Jul 2025
Sensitivity Analysis: Negative Control (Predicting 2018-2019)
Positivity Rate
0%
10%
20%
30%
Observed
Predicted (Hold-out)
Jan 2018
Jul 2018
Jan 2019
Jul 2019
Date